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Number:GJK24050177

Inspection Report

Sample Name: SHM-XE Intelligent Operation Box for Tap Switches

Contracting Organization: Shanghai Huaming Electric Power Equipment Manufacturing Co., Ltd.

Manufacturer: Shanghai Huaming Electric Power Equipment Manufacturing Co.,Ltd.

Inspection Category: Type Test

National Quality Inspection and Testing Center for Power Transmission and Distribution Safety Control Equipment; Experimental Verification Center of State Grid Electric Power Research Institute Co., Ltd.

August 7, 2024

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General Report Table for Inspection Results

Kind of inspection	<i>Type Testing</i>	Sample Model	SHM-XE
		Sample Name	Intelligent Operation Mechanism Box for Tap Switches
Client	Shanghai Huaming Electric Power Equipment Manufacturing Co., Ltd.	Address of the sponsoring organization	No.977, Tongpu Road, Putuo District, Shanghai
Production Unit	Shanghai Huaming Electric Power Equipment Manufacturing Co., Ltd.	Manufacturing Unit Address	No.977, Tongpu Road, Putuo District, Shanghai
Sample size	1 unit	Sample Number	GJK24050177
Sample Receipt Date	July 1, 2024	Sample Reception Status	Well-tended appearance
Software Version No.	V1.0	Software Verification Code	/
Proving time	July 4, 2024 to July 29, 2024		
Place of survey	No.19, Chengxin Avenue, Jiangning District, Nanjing City		
Inspection standard	GB/T13729-2019 Remote Control Terminal Equipment DL/T1498.1-2016 Technical Specifications for Online Monitoring Devices of Substation Equipment— Part1: General Requirements; Technical Specification for the Intelligent Operation Mechanism Box of SHM-XE Tap Switch (Judgment Criteria) GB/T17626.2-2018 Electromagnetic Compatibility Testing and Measurement Techniques – Test for Susceptibility to Electrostatic Discharge Interference GB/T 17626.3-2016 Electromagnetic Compatibility Testing and Measurement Techniques – Test for Resistance to Radio Frequency Electromagnetic Field Radiation Interference GB/T 17626.4-2018 Electromagnetic Compatibility Testing and Measurement Techniques – Anti-interference Test for Rapid Transient Pulse Sequences GB/T 17626.5-2019 Electromagnetic Compatibility Testing and Measurement Techniques – Surge (Impact) Resistance Test GB/T 17626.6-2017 Electromagnetic Compatibility Testing and Measurement Techniques – Conductive Interference Resistance to Radio Frequency Field Induction GB/T 17626.8-2006 Electromagnetic Compatibility Testing and Measurement Techniques – Test for Resistance to Interference from Power Frequency Magnetic Fields GB/T 17626.9-2011 Electromagnetic Compatibility Testing and Measurement Techniques – Pulse Magnetic Field Disturbance Resistance Test GB/T 17626.10-2017 Electromagnetic Compatibility Testing and Measurement Techniques – Test for Susceptibility to Magnetic Interference in Damped Oscillations GB/T 17626.11-2008 Electromagnetic Compatibility Testing and Measurement Techniques – Anti-interference Test for Voltage Sags, Short-term Interruptions, and Voltage Changes		
Conclusion	Sample from Shanghai Huaming Electric Power Equipment Manufacturing Co., Ltd. the SHM-XE tap changer intelligent operating mechanism box has passed the test, and the results meeting all the requirements. This report consists of two parts: Part 1 – Function and Performance; Part 2 – Electromagnetic Compatibility.		
Sign and issue	Wan Dechun 万德春	Date of issue	August 1, 2024
Remarks	/		

Sample Description

1. Sample Photo



2. Sample Overview

Sample Name	Configuration Details	Product Serial No.	Manufacturer
SHM-X Tap changer intelligent operating mechanism box	Power Supply: AC 380 V, 50/60 Hz; Interface Configuration: RS485 Interface: 2 channels Analog Input Interface:Route 4 Switched Output Interface:Route 20 Switched Input Interface:Route 9 Analog Output Interface:Route 2 Network Ports: 2	SHM-XE-2024	Shanghai Huaming Electric Power Equipment Man u-facturing Co., Ltd.

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Part 1: Functions and Performance

Inspection Report Sub-Table

Sample Model	SHM-XE	Sample Name	Intelligent Operation Mechanism Box for Tap Switches		
Sample size	1 Table	Sample Number	GJK24050177		
Sample Receipt Date	July 1, 2024	Sample Reception Status	Well-tended appearance		
Software Version No.	V1.0	Software Verification Code	/		
Proving time	July 4, 2024 to July 29,2024				
Place of survey	No.19, Chengxin Avenue, Jiangning District, Nanjing City				
Inspection standard	GB/T 13729-2019 Remote Control Terminal Equipment DL/T 1498.1-2016 Technical Specifications for Online Monitoring Devices for Substation Equipment – Part 1: General Principles Technical Specification for the Intelligent Operating Mechanism Box of the SHM-XE Tap changer (Judgment Criteria)				
Main testing instruments and equipment: Name, Model, Number, and Validity Period					
Step-in environmental test chamber (15 cubic meters)		VCZ 50015-S	PAL/P-HJ. 0010	2024. 09. 06	
Step-type dust test chamber		SC-068	PAL/P-HJ. 0020	2024. 09. 06	
Water Spraying Test Device		JL-I/II	PAL/P-HJ. 0022	2024. 09. 06	
Insulation Voltage Test Instrument		3153	PAL/P-HJ. 0031	2024. 09. 06	
Shock Voltage Tester		PSURGE8000+PIM100+VTM15000	PAL/J-01. 0005	2024. 09. 06	
Electrodynamic vibration generator system		ES-100-550	PAL/P-JX. 0001	2024. 09. 06	
Single-phase AC/DC Standard Source		DK-51E1	PAL/D-09. 0008	2024. 09. 06	
Conclusion	Sample from Shanghai Huaming Electric Power Equipment Manufacturing Co., Ltd.: the SHM-XE tap changer intelligent operating mechanism box. Upon inspection—including structural and visual examination, basic functionality tests, and measurement accuracy evaluations—the results fully comply with all specified requirements.				
Chief Inspector	The guest is	Night is...	Proofread	Li Xi	Examine and verify Zhang Xin
Remarks	/				

List of Inspection Items

Order No.	Test item		Conclusion	Remarks
1	Structural and Visual Inspection		Qualified	
2	Basic Function		Qualified	
3	Insulating property	Insulation resistance	Qualified	
		Dielectric strength	Qualified	
		Impulse voltage	Qualified	
4	Environmental suitability	Low temperature	Qualified	
		High temperature	Qualified	
		Alternating moist and hot conditions for 2 days (48 hours)	Qualified	
		Constant moist heat for 2 days (48 hours)	Qualified	
5	Mechanical behavior	Vibration Durability	Qualified	
		Impact Durability	Qualified	
		Collide	Qualified	
6	External Shell Protection	Dustproof (IP5X)	Qualified	
		Waterproof (IPX5)	Qualified	
7	Measurement error		Qualified	

1 Structural and Visual Inspection

Date: 2024.7.4

Temperature: 20°C Relative humidity: 55%

Technical Requirements: The device chassis shall incorporate necessary measures to prevent electromagnetic interference; its exposed conductive parts shall be electrically integrated and reliably grounded; the chassis must meet the ventilation and heat dissipation requirements of heat-generating components; chassis modules shall allow for easy insertion and removal with reliable contact and good interchangeability; the exterior coating or electroplating layer shall be firm, uniform, smooth, and free from peeling or corrosion.

inspection result :

Appeal- inspec- tion	Scope of examination	Result
	Tamper-proof	√
	Electrical wiring	√
	Ventilation and heat dissipation	/
	Case Module	/
	Surface coating	√

Note: The symbol "" √ "" indicates that the test results meet technical requirements, while "/" signifies inapplicability.

Conclusion: **Qualified****2 Basic Functions**

Date: July 4–5, 2024

Temperature: 20°C Relative humidity: 55%

Technical requirements: The device under test must possess the functions listed in the table below.

Actual Test Results:

Order number	Technical requirement		Detection result
1	Online Monitoring System	Real-time monitoring of the transmission gear position information of the operating mechanism should be supported.	√
2		Real-time monitoring of motor status should be supported	√
3		Real-time monitoring of critical parameters such as switching vibration, oil temperature, and oil pressure of tap changers should be supported.	√
4	Instructions and Records Function	The display should support showing tap changer position information and operation counts.	√
5		Support the tap changer code disk pointer indication	√
6		Support operation counts, log information, and fault record tracking	√
7	Analog Data Acquisition Collection and Output Functions	Support should be provided for measuring the AC input voltage and AC input current of transformers and tap changers.	√
8		Support output of analog signals such as 0–20 mA, 4–20 mA, ±1 mA, and 1–5 V.	√
9	Defensive function	Dual protection using both electrical limit devices and mechanical limit devices should be implemented.	√
10		Motor overload protection and locking functions should be supported.	√
11		The monitoring and locking protection function of optical CT circuit breakers should be supported.	√
12		The system should support power-off memory and power-on recovery functions. In the event of an abnormal power outage during tap-changing, the process will resume after restart to complete any pending tap-changing operations.	√
13		It should support real-time monitoring of power supply conditions such as phase loss or phase sequence errors in three-phase systems, and implement motor operation lockout functionality.	√
14		The lockout functions for tap changer timeout and incomplete switching should be supported.	√
15	Operation Control Function	It should support three operational modes: remote control, local operation, and numerical control.	√
16		Support basic operations such as raising, lowering, and stopping at the local level.	√
17		Support progressive operations to perform multi-level continuous tasks	√

Order number	Technical requirement		Detection result
18	Contact Configuration Function	Users should be able to customize the configuration of operation status output contacts	√
19		Users should be able to customize the input contacts for overcurrent lockout protection.	√
20		Users should be able to customize the configuration of BCD code position signal output contacts.	√
21		Users should be able to customize the configuration of location signal output contacts	√
22	Maintenance and Care Functions	The system should support determining contact wear severity and predicting the lifespan of other components by analyzing the number of switching operations and recorded action durations, thereby providing recommendations for maintenance, upkeep, and repair.	√

Note: The symbol "√" indicates that the test results meet the technical requirements.

Conclusion: **Qualified**

3 Insulation Performance

Date: July 5, 2024 temperature : 20°C Relative humidity: 55%

3.1 Insulation Resistance

Technical requirement: The insulation resistance of the power circuit to ground shall be no less than 100 MΩ.

Measured results: see the "Insulation Performance Test Table" for details.

Conclusion: **Qualified**

3.2 Medium Strength

Technical requirements: The power circuit must withstand an AC withstand voltage test of 2000 V at industrial frequency for 1 minute, with no breakdown, flashover, or component damage observed.

Measured results: see the "Insulation Performance Test Table" for details.

Conclusion: **Qualified**

3.3 Impact Voltage

Technical requirements: The power circuit must withstand a short-duration impulse voltage test using a standard lightning wave with an open-circuit test voltage of 5 kV and a duration of 1.2/50 μs; the device shall not exhibit insulation breakdown, flashover, or damage.

Measured results: see the "Insulation Performance Test Table" for details.

Conclusion: **Qualified**

Insulation Performance Test Table

Test Area	Insulation resistance (MΩ)	Dielectric strength		Impulse voltage	
		Voltage (kV)	Result	Voltage (kV)	Result
The power supply circuit is connected to ground.	>500	2.0	√	5.0	√
Switched-input signal relative to ground	>500	0.5	√	1.0	√
Switched-output signal to ground	>500	0.5	√	1.0	√
The RS-232 interface is connected to ground.	>500	0.5	√	1.0	√
The RS-485 interface is connected to ground.	>500	0.5	√	1.0	√
The input voltage of the AC circuit is connected to ground.	>500	2.0	√	5.0	√
The alternating current input is directed towards the ground.	>500	0.5	√	1.0	√

Note: The symbol "√" indicates that the test results meet the technical requirements.

4 Environmental Adaptability

4.1 Low Temperature

Date: July 5, 2024 temperature : 20°C Relative humidity: 55%

Technical requirements: The test device shall be placed in the environmental test chamber while powered on. Under conditions where the temperature deviation does not exceed $\pm 2^{\circ}\text{C}$, the device shall be cooled at a rate of no more than $1^{\circ}\text{C}/\text{min}$ until the temperature reaches -40°C and stabilizes; timing shall then commence for a duration of 2 hours. During and after the test, the device shall function properly.

Actual Test Results:

Ambient condition	Proving time	Technical requirement	Result
-40°C	2h	During and after the trial period, the device shall operate normally, and its network communication functions shall function properly.	√

Note: The symbol "√" indicates that the test results meet the technical requirements.

Conclusion of this item: **Qualified**

4.2 High Temperature

Date: July 5, 2024 temperature : 20°C Relative humidity: 55%

Technical requirements: The test device shall be placed in the environmental test chamber while powered on. Under conditions where the temperature deviation does not exceed $\pm 2^{\circ}\text{C}$, the temperature shall be increased at a rate not exceeding $1^{\circ}\text{C}/\text{min}$ until it reaches 70°C and stabilizes; timing shall then commence and continue for 2 hours. During and after the test, the device shall function properly.

Actual Test Results:

Ambient condition	Proving time	Technical requirement	Result
70°C	2h	During and after the trial period, the device shall function properly and its network communication capabilities shall remain normal.	√

Note: The symbol "√" indicates that the test results meet the technical requirements.

Conclusion: **Qualified**

4.3 Constant Humidity and Temperature

Date: July 8, 2024 – July 10, 2024 Temperature: 20°C Relative Humidity: 55%

Technical requirements: The device shall withstand a constant hot and humid test at a temperature of $(40 \pm 2)^{\circ}\text{C}$ and relative humidity of $(93 \pm 3)\%$ for a duration of 48 hours; during and after the test, the device must remain powered and function properly.

Actual Test Results:

Ambient condition	Proving time	Technical requirement	Result
Temperature: 40°C ; Relative humidity: 93%	48h	During and after the trial period, the device shall function properly, and its network communication capabilities shall remain normal.	√

Note: The symbol "√" indicates that the test results meet the technical requirements.

Conclusion: **Qualified**

4.4 Alternating Humidity and Heat

Date: July 5, 2024 – July 7, 2024 Temperature: 20°C Relative Humidity: 55%

Technical requirements: The device shall withstand an alternating wet-heat test at high temperature of 55°C , low temperature of 25°C , relative humidity of 95%, and a cycle count of 2 cycles. During and after the test period, the device must be powered on and function properly.

Actual Test Results:

Ambient condition	Number of cycles	Technical requirement	Result
High temperature: 55°C; Low temperature: 25°C; Relative humidity: 95%.	2	During and after the trial period, the device shall operate normally, and its network communication functions shall function properly.	√

Note: The symbol "√" indicates that the test results meet the technical requirements.

Conclusion:**Qualified**

5 Mechanical Properties

Date: July 18,2024 – July 23,2024 Temperature: 20°C Relative humidity: 55%

5.1 Vibration Durability

Technical Requirements: Frequency range: 10 Hz to 150 Hz; Acceleration amplitude: 10.0 m/s²; Time per frequency sweep cycle: 8 minutes; Per axis

Number of frequency sweep cycles along the line direction: 20 Total duration of tests across three mutually perpendicular axes: 480 min. After testing, the device should not exhibit any issues such as loose fasteners or mechanical damage.

Test results: After the test, no loosening of fasteners or mechanical damage was observed in the device.

Conclusion of this item:**Qualified**

5.2 Impact Durability

Technical requirements: Acceleration amplitude: 150 m/s²; Pulse duration: 11 ms; Number of pulses per direction: 3; Number of pulses per axis: 6; Total number of pulses across all three axes: 18.

After testing, the device should not exhibit any issues such as loose fasteners or mechanical damage.

Test results: After the test, no loosening of fasteners or mechanical damage was observed in the device.

Conclusion:**Qualified**

5.3 Collision

Technical requirements: Acceleration amplitude: 100 m/s²; Pulse duration: 16 ms; Number of collisions per direction: 1000; Number of collisions per axis: 2000; Total number of collisions across all three axes: 6000.

After testing, the device should not exhibit any issues such as loose fasteners or mechanical damage.

Test results: After the test, no loosening of fasteners or mechanical damage was observed in the device.

Conclusion of this item:**Qualified**

6. Housing Protection

Date: July 25,2024 – July 29,2024 Temperature: 20°C Relative humidity: 55%

6. 1 Dustproof (IP6X)

Technical requirements: The device shall be placed in the dust-proof test chamber at its normal operating position, with talcum powder dosage of 2 kg per cubic meter of chamber volume; the internal pressure of the device housing shall be maintained below atmospheric pressure using a vacuum pump, with a maximum pressure differential of 2 kPa and a pumping rate lower than 40 times the volume of the housing per hour; after 8 hours of testing, the device shall be opened for inspection to ensure no dust deposition is present.

Test results: The device's housing meets IP6X dust protection requirements; no visible dust was detected inside after the test, and it operates safely and normally when powered on.

Conclusion of this item:**Qualified**

6.2 Water Resistance (IPX6)

Technical requirements: The device shall be mounted on a continuously rotating turntable; the nozzle has an inner diameter of 12.5 mm and a water flow rate of (100 ± 5) L/min; the distance from the nozzle to the outer shell surface ranges from 2.5 m to 3 m; the water spray duration per square meter of the outer shell is approximately 1 minute; the total test duration shall not be less than 3 minutes; and the water spray test must not cause any harmful effects to the device.

Test results: The device's housing meets IPX6 waterproofing requirements; the water spray test caused no adverse effects, and the device operated normally upon power-on.

Conclusion:Qualified

7 Measurement Error

7.1 AC Voltage

Date: July 4, 2024 Temperature: 20°C Relative Humidity: 55%

Technical requirement: The voltage measurement error shall not exceed ±0.5%.

Actual Test Results:

Channel	Input voltage	Nominal Voltage (V)	Terminal Measurement Value (V)	Error
Transformer Monitoring	100%	380. 000	379. 77	-0. 06%
	80%	304. 000	303. 79	-0. 06%
	60%	228. 000	227. 77	-0. 06%
	40%	152. 000	151. 80	-0. 05%
	20%	76. 000	75. 96	-0. 01%
	0%	0. 000	0. 00	0. 00%
Tapping switch monitoring	100%	380. 000	380. 15	0. 04%
	80%	304. 000	304. 11	0. 03%
	60%	228. 000	228. 21	0. 06%
	40%	152. 000	152. 07	0. 02%
	20%	76. 000	76. 07	0. 02%
	0%	0. 000	0. 00	0. 00%

Conclusion: Qualified

7.2 Alternating Current

Date:2024.7. Temperature: 20°C Relative humidity: 55%

Technical requirement: The current measurement error shall not exceed ±0.5%.

Actual Test Results:

Channel	Input currenton	Current Nominal Value (A)	Terminal Measurement Value (A)	Error
Transformer Monitoring	100%	10. 0000	10. 000	0. 00%
	80%	8. 0000	7. 998	-0. 02%
	60%	6. 0000	5. 996	-0. 04%
	40%	4. 0000	3. 996	-0. 04%
	20%	2. 0000	1. 999	-0. 01%
	0%	0. 0000	0. 000	0. 00%

Channel	Input currenton	Current Nominal Value (A)	Terminal Measurement Value (A)	Error
Tapping switch monitoring	100%	10.0000	9.997	-0.03%
	80%	8.0000	7.998	-0.02%
	60%	6.0000	6.000	0.00%
	40%	4.0000	3.999	-0.01%
	20%	2.0000	1.999	-0.01%
	0%	0.0000	0.000	0.00%

Conclusion: Qualified

No further content follows in this section

Part Two: Electromagnetic Compatibility

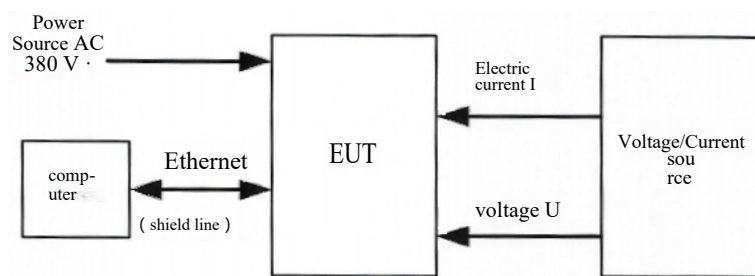
Inspection Report Sub-Table

Sample Model	SHM-XE	Sample Name	Intelligent Operation Mechanism Box for Tap Switches		
Sample size	1 unit	Sample Number	GJK24050177		
Sample Receipt Date	July 11,2024	Sample Reception Status	Well-tended appearance		
Software Version Number	/	Software Verification Code	/		
Proving time	July 15,2024 – July 16,2024				
Place of survey	No.19, Chengxin Avenue, Jiangning District, Nanjing City				
Inspection standard	GB/T 17626.2-2018 Electromagnetic Compatibility Testing and Measurement Techniques – Electrostatic Discharge Susceptibility Test GR/T 17626.3-2016 Electromagnetic Compatibility Testing and Measurement Techniques – Test for Resistance to Radio Frequency Electromagnetic Field Radiation Interference GB/T 17626.4-2018 Electromagnetic Compatibility Testing and Measurement Techniques – Anti-interference Test for Rapid Transient Pulse Sequences GB/T 17626.5-2019 Magnetic Compatibility Testing and Measurement Techniques – Surge (Impact) Resistance Test GB/T 17626.6-2017: Test and Measurement Techniques for Magnetic Compatibility – Resistance to Conductive Interference from Radio Frequency Field Induction GB/T 17626.8-2006 Electromagnetic Compatibility Testing and Measurement Techniques – Test for Resistance to Interference from Power Frequency Magnetic Fields GB/T 17626.9-2011 Electromagnetic Compatibility Testing and Measurement Techniques – Pulse Magnetic Field Disturbance Resistance Test GB/T 17626.10-2017 Electromagnetic Compatibility Testing and Measurement Techniques – Test for Susceptibility to Magnetic Interference in Damped Oscillations GB/T 17626.11-2008 Electromagnetic Compatibility Testing and Measurement Techniques Noise immunity tests for voltage sags, short-term interruptions, and voltage fluctuations – DL/T 1498.1-2016 Technical specifications for online monitoring of substation equipment – Part 1: General requirements				
Main testing instruments and equipment: Name, Model, Number, and Validity Period					
High-voltage electrostatic discharge generator; Surge generator	ONYX30	PAL/F-02. 0004	2024. 10. 12		
	VCS500N10	PAL/F-05. 0002	2025. 02. 27		
Electric Rapid Transient Pulse Cluster Simulator – Damped Oscillation Wave Generator	EFT500N8	PAL/F-04. 0001	2025. 02. 27		
	OCS500N6. 8 15003	PAL/F-18. 0003	2025. 02. 27		
Three-phase harmonic testing system; Conductive immunity generator; Radiation signal generator; Power amplifier	ix-CTS/EOS-3 CWS	PAL/F-11. 0002	2024. 10. 11		
	500N1. 4 SMB100A	PAL/F-11. 0002	2024. 10. 11		
	BBA100-B1000C800	PAL/F-03. 0003	2025. 02. 27		
	STLP9129	PAL/F-03. 0008	2024. 11. 09		
Logarithmic-period antenna power-frequency magnetic field generator	PMM1008	PAL/F-03. 0028	2025. 02. 27		
		PAL/F-08. 0003	2025. 02. 27		
Conclusion	Sample from Shanghai Huaming Electric Power Equipment Manufacturing Co., Ltd.: the SHM-XE tap changer intelligent operating mechanism box. Test results for the following parameters met all required specifications: electrostatic discharge immunity, RF electromagnetic field radiation immunity, immunity to electrical fast transient pulse clusters, surge (impulse) immunity, conductive interference immunity due to RF field induction, power frequency magnetic field immunity, impulse magnetic field immunity, damping oscillation magnetic field immunity, voltage sags, and short-term voltage interruptions.				
Chief Inspector	Chen Yangyang 陈扬扬	Proofread	Shen Xuemei 沈雪梅	Examine and verify	Xiao Baoming 肖保明
Remarks	/				

List of Inspection Items

Order No.	Inspecting item		Conclusion	Remarks
1	Electromagnetic compatibility	Electrostatic Discharge Immunity	Qualified	
2		Resistance to interference from radio frequency electromagnetic field radiation	Qualified	
3		Resistance to interference from groups of electric fast transient pulses	Qualified	
4		Surge (Impact) Resistance	Qualified	
5		Resistance to conducted interference caused by radio frequency field induction	Qualified	
6		Resistance to interference from power-frequency magnetic fields	Qualified	
7		Pulse magnetic field immunity	Qualified	
8		Damping oscillation magnetic field immunity to interference	Qualified	
9		Voltage sags and short-term interruptions	Qualified	

8 Test Device Connection Diagram



9 Performance Criteria

- During the testing process, if the functions or performance remain within the specified technical limits, it can be classified as Grade A.
- During the test, if a function or performance temporarily declines or is lost but can recover spontaneously, it may be classified as Grade B.
- During the test procedure, if a function or performance temporarily declines or is lost but requires manual intervention or system reset to be restored, it can be classified as Grade C.
- During the testing process, if equipment (components) or software experiences damage (loss of stored data), loss of functionality, or performance degradation that cannot be restored even with manual intervention or system reset, it shall be classified as Level D.

10. Resistance to Electrostatic Discharge Interference

Date: 2024.07.16 temperature : 27°C Relative humidity: 49%

Testing Basis: GB/T 17626.2-2018 Electromagnetic Compatibility Testing and Measurement Techniques – Electrostatic Discharge Susceptibility Test

Technical Requirements: Meet the performance criteria of Grade B or Grade A.

Test Level: Level 4

Test values: Air discharge: ± 2 kV, ± 4 kV, ± 8 kV, ± 15 kV;

Contact discharge: ± 2 kV, ± 4 kV, ± 6 kV, ± 8 kV.

Test Method: Direct Discharge: With the test device in normal operating condition, apply stepwise air discharge voltage and contact discharge voltage to both non-metallic and metallic parts of the panel that are easily accessible to the hand. The number of discharge cycles for both positive and negative terminals at each test point shall exceed 10 times; observe the operational status of the test device.

testing arrangement :



Actual Test Results:

Discharge Form	Direct Discharge	
	Contact discharge	Air discharge
Result	Normal	Normal

Conclusion: **Qualified**

11. Resistance to Interference from Radio Frequency Electromagnetic Field Radiation

Date: 2024.07.15 temperature : 27℃ Relative humidity: 49%

Testing Basis: GB/T 17626.3-2016 Electromagnetic Compatibility – Testing and Measurement Techniques – Test for Resistance to Radio Frequency Electromagnetic Field Radiation Interference

Technical Requirements: Meet the Class A performance criteria.

Test Level: Level 3

Test value: Test level: 10 V/m;

 Frequency range: 80 MHz to 1 GHz;

 Modulation:80% AM;

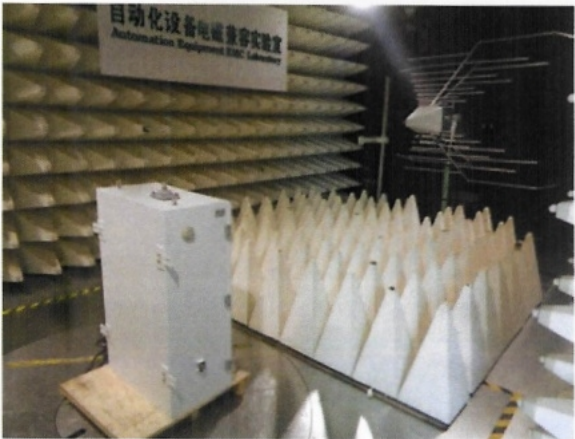
 Modulation Frequency: 1 kHz

 Automatic scan frequency interval: 1% of the previous frequency;

 Duration for each frequency point: 0.5s

Test method: Place the test device in a uniform field of 10 V/m and observe its operational performance.

testing arrangement :



Actual Test Results:

Antenna polarization direction	Horizontal direction		Vertical direction	
Device Direction	Previous	Behind	Previous	Behind
Result	In working order	In working order	In working order	In working order
Meet performance criteria	Class A	Class A	Class A	Class A

Conclusion: Qualified

13. Surge (Impact) Resistance

Date: 2024.07.16 temperature : 27°C Relative humidity: 49%

Test Basis: GB/T 17626.5-2019 Electromagnetic Compatibility Testing and Measurement Techniques – Surge (Impact) Resistance Test

Technical requirements: Meet the performance criteria of Grade B or Grade A.

Test Level: Level 4

Test Value: Test Levels: Line-to-ground ±0.5 kV, ±1 kV, ±2 kV, ±4 kV

Line voltage: ±0.5 kV, ±1 kV, or ±2 kV;
Number of test pulses: 5 positive and 5 negative pulses each.

Pulse interval: 60s.

Test Method: The test device is operated under normal conditions. According to the specified test values required by the test category, interference signals are sequentially applied to the power supply and control/signal port circuits via a coupling/decoupling network, and the operational status of the device is observed.

testing arrangement :



Actual Test Results:

Test Port		Source	Voltage	Current	Enter	Open
Result	Line-to-ground	Normal	Normal	Normal	Normal	Normal
	Line-Line	Normal	Normal	Normal	Normal	Normal
Meet performance criteria		Class A	Class A	Class A	Class A	Class A

Conclusion: Qualified

14. Resistance to Conducted Interference Caused by Radio Frequency Field Induction

Date: 2024.07.16 temperature : 27°C Relative humidity: 49%

Testing Basis: GB/T 17626.6-2017 Electromagnetic Compatibility – Testing and Measurement Techniques – Resistance to Conductive Interference from Radio Frequency Field Induction

Technical Requirements: Meet the Class A performance criteria.

Test Level: Level 3

Test value: Test level: 10 V;

 Scan range: 150 kHz to 80 MHz;

 Amplitude Modulation: 80% AM;

 Modulation frequency: 1 kHz;

 Step size: 1% of the previous frequency point;

 Stay time: 0.5s.

Test Method: The test was conducted in a shielded chamber with the device under test operating under normal conditions. Interference signals were applied to the power supply and control/signal ports of the device via a coupling/decoupling network, and delivered to the Ethernet communication port through an electromagnetic clamp, followed by observation of the device's operational performance.

testing arrangement :



Actual Test Results:

Test Port	Source	Voltage	Current	Enter	Open	Ethernet Port
Result	Normal	Normal	Normal	Normal	Normal	Normal
Meet performance criteria	Class A	Class A	Class A	Class A	Class A	Class A

Conclusion: Qualified

15. Resistance to Magnetic Field Interference at Power Frequency

Date: 2024.07.16 temperature : 27°C Relative humidity: 49%

Testing Basis: GB/T 17626.8-2006 Electromagnetic Compatibility – Testing and Measurement Techniques – Test for Susceptibility to Power Frequency Magnetic Fields

Technical Requirements: Meet the Class A performance criteria.

Test Level: Level 5

Test value: Stable magnetic field: 100 A/m;

Short-term magnetic field: 1000 A/m.

Test Method: The test device is operated under normal conditions. In accordance with the specified test values for each test category, the proximity method is employed to apply interference signals to the test device along three mutually perpendicular directions, and the operational status of the device is observed.

testing arrangement :



Actual Test Results:

Magnetic Field Type	Steady state magnetic field			Short-term magnetic field		
	X direction	Y direction	Z direction	X direction	Y direction	Z direction
Result	In working order	In working order	In working order	In working order	In working order	In working order
Meet performance criteria	Class A	Class A	Class A	Class A	Class A	Class A

Conclusion: Qualified

16 Pulse Magnetic Field Antiplague Resistance

Date: 2024.07.16 temperature : 27°C Relative humidity: 49%

Testing Basis: GB/T 17626.9-2011 Electromagnetic Compatibility Testing and Measurement Techniques – Pulse
Magnetic Field Disturbance Resistance Test

Technical Requirements: Meet the Class A performance criteria.

Test Level: Level 5

Test value: Test level: 1000 A/m (peak)

Number of pulses: 5 for each polarity (positive and negative);

Time interval: 10s.

Test Method: The test device is operated under normal conditions. According to the specified test values for each test category, the proximity method is employed to apply interference signals to the device in three mutually perpendicular directions. At least five positive-polarity pulses and five negative-polarity pulses are applied, with a 10-second interval between each pulse, while monitoring the device's operational performance.

testing arrangement :



Actual Test Results:

Magnetic Field Type	Positive pulse			Negative pulse		
	X direction	Y direction	Z direction	X direction	Y direction	Z direction
Result	In working order	In working order	In working order	In working order	In working order	In working order
Meet performance criteria	Class A	Class A	Class A	Class A	Class A	Class A

Conclusion: Qualified

17. Resistance to Magnetic Field Interference in Damped Oscillations

Relative humidity: 49%

Test Basis: GB/T 17626.10-2017 Electromagnetic Compatibility – Testing and Measurement Techniques – Test for susceptibility to damping oscillating magnetic fields

Technical Requirements: Meet the Class A performance criteria.

Test Level: Level 5

Test value: 100 A/m;

Test Method: The test device is operated under normal conditions. According to the specified test values for each test category, the proximity method is employed to apply damping oscillating magnetic field interference signals at 0.1 MHz and 1 MHz to the test device in three mutually perpendicular directions, and the operational status of the device is observed.

testing arrangement :



Actual Test Results:

Magnetic Field Type	0.1 MHz magnetic field			1 MHz magnetic field		
	X direction	Y direction	Z direction	X direction	Y direction	Z direction
Result	In working order	In working order	In working order	In working order	In working order	In working order
Meet performance criteria	Class A	Class A	Class A	Class A	Class A	Class A

Conclusion: Qualified

18 Voltage Sags and Short-Term Interruptions

Date: 2024.07.16 Temperature: 27°C | Relative humidity: 49%

Testing Basis: GB/T 17626.11-2008 Electromagnetic Compatibility Testing and Measurement Techniques

Technical requirements for immunity testing against voltage sags, short-term interruptions, and voltage fluctuations: Must meet Performance Criterion Level B or Level A requirements.

Test Grade: Class 3

Test values:	Voltage sag:80%; Duration: 5000 ms	3 times
	70%; Duration:	1000ms 3 times
	40%; Duration:	200ms 3 times
	0%; Duration:	20ms 3 times
	0%; Duration:	10ms 3 times

Test Method: The test device shall be operated under normal conditions. According to the specified test values required by the test category, a short-term voltage sag shall be applied to the working power supply of the test device, and its ability to function normally shall be observed.

testing arrangement :



Actual Test Results:

Test item	Source				
	Voltage sag				
	80%/5000ms	70%/1000ms	40%/200ms	0%/20ms	0%/10ms
Result	In working order	In working order	In working order	In working order	In working order
Meet performance criteria	Class A	Class A	Class A	Class A	Class A

Conclusion: Qualified

No further content follows in this section